

DAY 01 MILESTONE PRESENTATION

Green Skilling, AI & Sustainability

Shell-Edunet Skills4Future AICTE Internship program. Navigating eco-conscious technology pipelines, real-time climate initiatives, and data metrics.

NAME

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ENROLLMENT

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STREAM

Information Technology

COLLEGE

RNG Patel Institute of Technology

01 / Core Concepts

Unveiling the foundational principles of Ecological awareness,
Technological competence, and Sustainability paradigms.

Defining the Modern Green Shift

What is Green Skilling?

The structured acquisition of practical knowledge, technical values, and specialized abilities required to design, implement, and maintain low-carbon systems.

Fosters adaptive competence in climate regulation compliance and clean-tech engineering across emerging digital industries.

- Resource Optimization
- Renewable Integration
- Carbon Analysis

What is Sustainability?

An ongoing operational standard that focuses on meeting current generational requirements without compromising the resource capacity of tomorrow.

Maintains dynamic system stability by balancing continuous technological progress with deep-rooted environmental protection guidelines.

- Ecological Balance
- Economic Viability
- Social Governance

Key Insights & Resources Learned



SDG 17: Partnerships

Establishing collaborative pipelines between public institutions and private giants. The joint Shell and Edunet foundation initiative launches skill labs across state ITIs to scale clean energy knowledge.



Data Centers

Extracting data from documentative profiles on computational infrastructure. Highlighting the critical need to optimize cooling systems, decrease baseline electricity usage, and enforce water stewardship.



National Policies

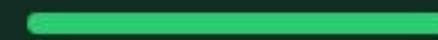
Navigating structural frameworks designed by national boards, such as the Skill Council for Green Jobs (SCGJ), targeted at standardizing professional certifications in solar, waste, and EV sectors.

The Dual Nature of Artificial Intelligence

DOMAIN AREA	THE GOOD (SUSTAINABILITY DRIVERS)	THE BAD (ENVIRONMENTAL FOOTPRINT)
Compute & Grids	AI optimizes smart grids and predicts load demands, preventing peak-hour fossil fuel reliance.	Heavy power and water consumption for training large language models (LLMs).
Biodiversity	Computer vision networks track endangered wildlife and mapping forest fragmentation.	Mineral extraction for hardware components degrades natural habitats.
Circular Economy	AI-powered sorting robots segregate complex waste categories with hyper-precision.	Model obsolescence accelerates computing hardware disposal and increases e-waste.

02 / Activities & Lab Metrics

A closer look at programmatic exploration, numeric simulations, and key carbon patterns captured during the internship sessions.



Jupyter Notebook Lab Activities

Analyzing local variables, computing indices, and compiling libraries to track footprint levels.

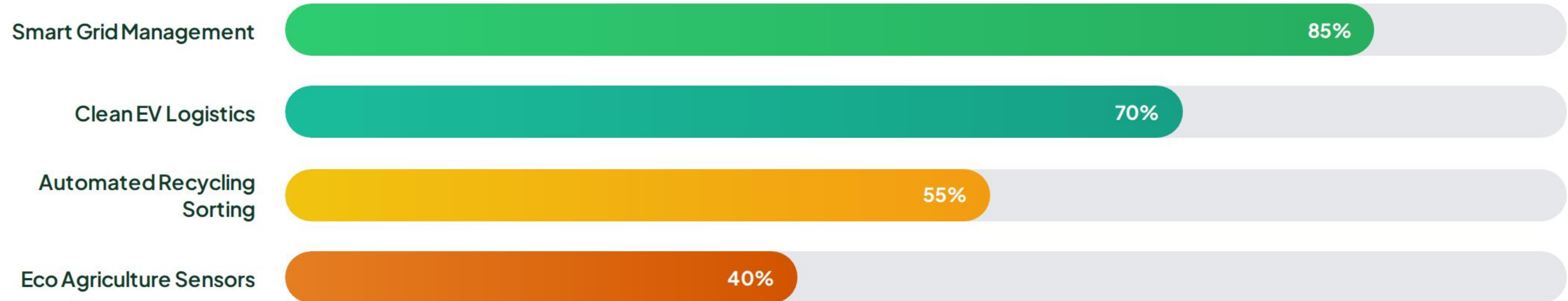
 **Mathematical Formulations:** Calculating total carbon output based on grid operational parameters using programmatic classes.

 **NumPy Array Foundations:** Utilizing dense multidimensional arrays to map out clean energy production levels across seasonal quarters.

 **Pandas Manipulation:** Processing CSV structures representing municipal electricity demand trends, applying advanced cleanup filters.

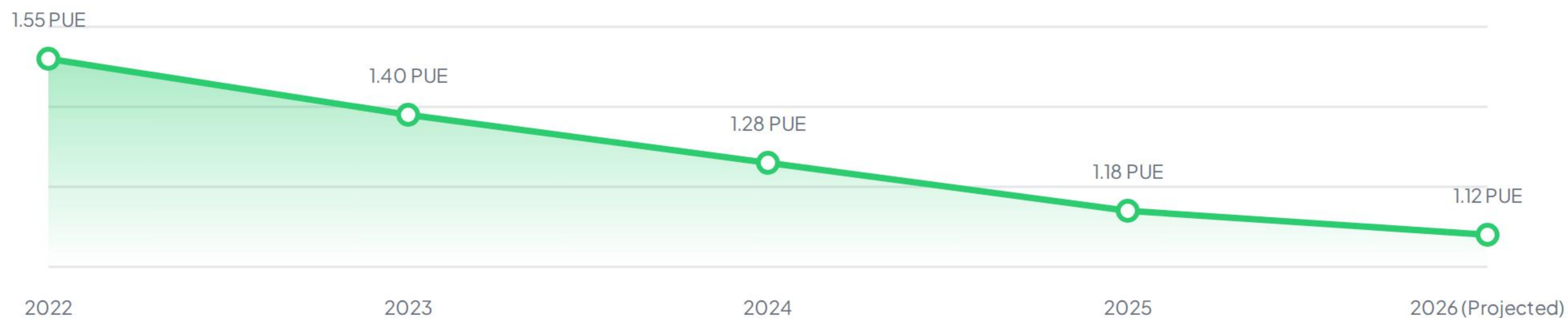
 Green technology dashboard showcasing clean code and energy data analytics on screen

Projected Sector Efficiency Improvements



Simulated impact margins achieved through algorithmic optimization. Smart grids display the highest capacity for waste prevention.

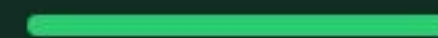
Optimized Data Center PUE Efficiency Curve



Visual representation of power efficiency index (PUE) scaling towards optimal ratio (1.00) through advanced AI workload leveling.

03 / Proposed Vision

Initiating the Python Capstone project designed to balance technical coding execution with real-world sustainability benchmarks.



Capstone Project Initiative

AI-Driven Smart Circular Waste Classifier

Deploying deep learning pipelines to dynamically parse, label, and route organic, recyclable, and chemical waste materials.

Technical Stack: Built on Python using OpenCV for image parsing and PyTorch CNNs for custom image classification. Directly aligns with SDG 12 (Responsible Consumption) and SDG 13 (Climate Action).



Questions?

Thank you for your valuable attention.

Patel Jeni Ashokkumar

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Image Sources



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